THE UNIVERSITY OF BAMENDA

THE COLLEGE OF TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

DESIGN AND IMPLEMENTATION OF A WEB-BASED WATER QUALITY MONITORING SYSTEM FOR LOCAL COMMUNITIES IN BAMENDA, CAMEROON

A Project Submitted to the Department of Computer Engineering in the College of Technology of the University of Bamenda in Partial Fulfillment of the Requirements for the Award of a Bachelor of Technology in Computer Networks and System Maintenance

BY:

Name

TEYIM YANICK NJIGMESHI

Registration Number:

UBA23PH049

SUPERVISOR:

Dr. ATANGANA OTELE

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ABSTRACT

Access to safe and clean water remains a major challenge in many local communities in Bamenda, Cameroon. This project presents the design and implementation of a web-based water quality monitoring system aimed at supporting local communities, councils, and non-governmental organizations (NGOs) in monitoring and managing water quality. The system allows community members to manually submit water quality observations and test results through a web application, creating a centralized and accessible platform for water quality data collection.

The application is developed using PHP (Laravel), HTML, CSS, and JavaScript, and provides features such as user authentication, water source registration, data visualization, and status reporting. Councils can verify the condition of water sources within their jurisdiction, while NGOs can use the collected data to provide recommendations for improving water quality. The system is designed to operate effectively under intermediate internet conditions common in Bamenda, with future support for sensor-based data integration beyond the scope of this project.

The proposed solution enhances community participation, improves access to water quality information, and supports informed decision-making to promote public health and environmental sustainability.

DEDICATION

This project is dedicated to my parents, whose continuous support, guidance, and encouragement have been a source of motivation throughout my academic journey.

ACKNOWLEDGEMENT

I sincerely express my gratitude to Central University Bamenda and all its staff for the knowledge and academic foundation provided throughout my studies. I also extend my sincere appreciation to my project supervisor for his guidance, encouragement, and valuable advice, which greatly contributed to the successful completion of this project.

My profound gratitude goes to my parents and family members for their moral, emotional, and financial support throughout my academic journey. I also thank my friends and classmates for their encouragement and assistance in one way or another. Above all, I thank the Almighty God for the gift of life, strength, and wisdom throughout the execution of this project.

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CHAPTER ONE

INTRODUCTION

1.1 Context (Background of the Study)

Access to safe and clean water is essential for human health, environmental sustainability, and socio-economic development. In many parts of Cameroon, particularly in the North West Region and the city of Bamenda, communities rely heavily on natural water sources such as streams, springs, wells, and community taps. These water sources are often vulnerable to contamination caused by poor sanitation practices, agricultural runoff, improper waste disposal, and limited environmental regulation.

In Bamenda, water quality monitoring is generally informal and irregular. Community members often rely on visual observation or personal experience to judge whether water is safe for consumption. Local councils and non-governmental organizations (NGOs), which play key roles in public health and environmental protection, lack a centralized system for collecting and analyzing water quality information across different communities.

With the growing availability of internet services at an intermediate level and the widespread use of digital technologies, web-based information systems offer a practical solution to this challenge. A web application can provide a centralized platform where water quality data is collected, stored, and accessed by relevant stakeholders. This project focuses on designing and implementing a web-based water quality monitoring system tailored to the needs of local communities in Bamenda, Cameroon.

1.2 Problem Statement

Despite the importance of clean water, water quality monitoring in Bamenda remains inadequate and unstructured. There is no centralized web-based platform that allows communities to systematically report water quality conditions or enables councils to verify and monitor water sources within their jurisdictions.

As a result, water quality data is often unavailable, fragmented, or outdated. Communities may continue to use contaminated water without timely awareness, increasing the risk of waterborne diseases. Local councils face challenges in making informed decisions due to the absence of reliable data, while NGOs struggle to identify priority areas for intervention.

Existing laboratory-based testing methods, although accurate, are expensive, time-consuming, and inaccessible to most local communities. Therefore, the key problem addressed by this study is the lack of a localized, affordable, and web-based system for collecting, managing, and monitoring water quality data in Bamenda.

1.3 Research Questions

1.3.1 General Research Question

How can a web-based application be used to improve water quality monitoring in local communities in Bamenda?

1.3.2 Specific Research Questions

- How can community members contribute water quality data through a web platform?
- How can councils verify water quality information within their jurisdictions?
- How can NGOs use collected data to make informed recommendations?

1.4 Objectives of the Study

1.4.1 General Objective

To design and implement a web-based water quality monitoring system that supports local communities, councils, and NGOs in monitoring water quality in Bamenda, Cameroon.

1.4.2 Specific Objectives

- To examine existing water quality monitoring practices in local communities in Bamenda
- To design a user-friendly web application for manual submission of water quality data
- To develop a centralized database for storing and managing water quality records

1.5 Justification and Motivation

The project is motivated by the need to improve access to water quality information using affordable and locally applicable web technologies.

1.6 Significance of the Study

The system will support communities, councils, and NGOs in monitoring water quality and improving public health outcomes.

1.7 Scope

The project focuses on manual data collection through a web application and excludes real-time sensor integration.

1.8 Limitations

One of the major limitations of this study is its dependence on internet connectivity, which may be inconsistent in some parts of Bamenda and surrounding communities, thereby affecting timely access to the web application. In addition, the system relies on manually submitted data from community members, which may vary in accuracy due to differences in users' knowledge, testing methods, or honesty. Since automated sensor-based data collection is not included in the current scope of the project, real-time and highly precise measurements cannot be guaranteed. These limitations may influence the completeness and reliability of the data collected; however, the system is designed to mitigate these challenges through data verification by local councils and structured data entry forms.

1.9 Delimitations

This study is deliberately limited in scope to Bamenda and nearby local communities in the North West Region of Cameroon in order to ensure relevance and manageability at the Bachelor level. The project focuses exclusively on the development of a web-based water quality monitoring application using manual data entry and does not include the design or implementation of physical water quality sensors or laboratory testing equipment. Furthermore, the system addresses water quality monitoring and reporting rather than direct water treatment or infrastructure development. These delimitations allow the study to concentrate on addressing local challenges using web technologies while providing a foundation for future expansion.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to water quality monitoring, web-based environmental information systems, and community-driven data collection approaches. The review establishes the theoretical and practical background of the study and highlights gaps in existing research that justify the need for a localized, web-based water quality monitoring system for communities in Bamenda, Cameroon.

2.2 Water Quality Monitoring

Water quality monitoring refers to the systematic process of collecting, analyzing, and interpreting data about the physical, chemical, and biological characteristics of water. According to the World Health Organization, regular monitoring of water sources is essential to ensure safe drinking water and to prevent waterborne diseases such as cholera, typhoid, and dysentery.

Traditional water quality monitoring methods rely heavily on manual sampling followed by laboratory analysis. Although these methods provide accurate results, they are often expensive, time-consuming, and inaccessible to rural and semi-urban communities. In many developing regions, including Cameroon, such approaches are not suitable for continuous monitoring due to limited resources and infrastructure.

As a result, there is growing interest in alternative monitoring approaches that are cost-effective, scalable, and capable of providing timely information to decision-makers and communities.

2.3 Web-Based Environmental Monitoring Systems

Web-based information systems have increasingly been adopted in environmental monitoring due to their ability to centralize data, enable remote access, and support real-time or near-real-time analysis. A web-based system allows users to collect and submit data through an internet-enabled platform, which is then stored in a centralized database and made accessible to authorized stakeholders.

Studies have shown that web applications improve data transparency and collaboration among users, authorities, and organizations. In the context of water quality monitoring, web-based platforms enable the aggregation of data from multiple locations, making it easier to identify trends, detect potential contamination, and support evidence-based decision-making.

The use of web technologies such as PHP frameworks, databases, and client-side scripting languages also enhances system scalability and maintainability. These technologies are particularly suitable for regions with intermediate internet connectivity, as they can be optimized for low bandwidth usage.

2.4 Community-Based Monitoring Approaches

Community-based monitoring involves the active participation of local residents in collecting and reporting environmental data. Several studies emphasize that involving communities in monitoring activities increases coverage, promotes environmental awareness, and encourages a sense of ownership over local resources.

In water quality management, community-driven data collection helps bridge the gap created by limited government resources. By allowing users to manually submit water quality observations or test results, authorities and organizations can gain access to localized and timely information.

However, effective community-based monitoring requires simple and user-friendly systems to ensure participation and data reliability. Web-based platforms provide an effective medium for supporting such initiatives by offering intuitive interfaces and centralized data management.

2.5 Role of Local Councils and NGOs in Water Management

Local councils play a crucial role in environmental health and water resource management within their jurisdictions. Their responsibilities include monitoring water sources, enforcing regulations, and responding to public health concerns. However, the lack of digital tools often limits their ability to perform these functions effectively.

Non-governmental organizations (NGOs) also contribute significantly to improving water quality through awareness campaigns, technical support, and intervention programs. Access to reliable data is essential for NGOs to identify priority areas and design effective solutions.

A web-based water quality monitoring system can support both councils and NGOs by providing verified and accessible data to guide planning, verification, and intervention efforts.

2.6 Gaps in Existing Studies

Although several studies have explored water quality monitoring systems, many focus on sensor-based or industrial-scale solutions that are costly and difficult to deploy in local African communities. Other studies emphasize mobile or IoT-based systems without considering internet limitations or the need for manual data entry.

There is limited research on web-based, community-driven water quality monitoring systems specifically tailored to the socio-economic and technological conditions of regions such as Bamenda. This gap highlights the need for a localized solution that combines community participation with web technology.

2.7 Summary of Literature Review

The literature reviewed demonstrates the importance of water quality monitoring and the growing role of web-based systems in environmental management. It also highlights the benefits of community participation in data collection and decision-making. However, existing solutions often fail to address the specific challenges faced by local communities in developing regions.

This study seeks to address these gaps by designing and implementing a web-based water quality monitoring system that is affordable, accessible, and suitable for local communities in Bamenda, Cameroon.

CHAPTER THREE

3.0 RESEARCH METHODOLOGY AND SYSTEM DESIGN

3.1 Introduction

This chapter presents the research methodology and system design for the Bamenda Water & Health Dashboard, a web-based water quality monitoring system for Mankon, Bamenda, Cameroon. It describes the research approach, requirements gathering methods, system architecture, database design, user interface design, development methodology, tools, testing strategy, deployment plan, and ethical considerations, translating stakeholder requirements into a concrete technical blueprint ready for implementation.

The study adopts a Design Science Research Methodology (DSRM), which focuses on creating and evaluating technological artifacts to solve real-world problems. This approach was selected because the primary outcome is a functioning web application addressing inadequate water quality monitoring in Mankon. DSRM consists of five sequential phases: problem identification and motivation from Chapter One, definition of solution objectives based on stakeholder needs, design and development as the focus of this chapter, demonstration through implementation, and evaluation through testing and user feedback. This methodology was chosen because it is problem-oriented, emphasizes rigorous evaluation, is widely accepted in computer science research, and provides clear structure appropriate for a bachelor's degree project

3.2 Justification for Design Science Approach

Alternative research methodologies were considered but found less suitable for this project. A pure survey methodology would describe existing conditions but would not produce a solution to improve them. An experimental methodology would test hypotheses under controlled conditions but would not address the specific contextual needs of Mankon. A case study approach would provide in-depth understanding of a single situation but would not result in a transferable artifact. The design science approach was therefore selected because it combines understanding of the problem domain with the creation of a practical solution, which aligns perfectly with the project's objectives of designing and implementing a functional web-based monitoring system.

3.3 Requirements Gathering

3.3.1 Data Collection Methods

Requirements were gathered through three complementary methods. Document analysis of NGO reports and Ministry publications revealed that water quality information exists only in scattered paper records that communities never see, confirming the need for a centralized digital platform. Stakeholder analysis identified primary actors including local residents, water point committee members, health workers, data entry clerks, and administrators, as well as secondary actors such as public health officials, NGO officers, researchers, and donors, with each actor's goals and pain points documented to derive functional requirements. Contextual inquiry revealed that community members already make sensory observations about water quality but lack a channel to report these findings to authorities.

3.3.2 Functional Requirements

Based on the requirements gathering activities, the following functional requirements were defined. The system must provide an interactive map showing all water sources in Mankon with color-coded pins indicating safety status. Users must be able to click on any pin to view detailed information about that water source including its name, location, type, test history, and community reports. Community members must be able to submit water quality observations through a simple mobile-friendly form that requires minimal typing. Authorized data entry personnel must be able to add official test results from laboratory analysis. Administrators must be able to moderate community reports, manage water source information, and manage user accounts. Health workers must be able to submit weekly disease case data for their facilities. All users must be able to view dashboard statistics and charts showing water quality trends.

3.3.3 Non-Functional Requirements

Non-functional requirements define quality attributes of the system. For performance, the page load time must be under three seconds on a 3G network connection, the map must load within five seconds, the API must respond within five hundred milliseconds, and the system must support fifty concurrent users. For usability, the system must be fully functional on smartphones with screens as small as 320 pixels, must work acceptably on low-bandwidth connections through optimized images and minimal JavaScript, and must follow accessibility guidelines. For security, all passwords must be hashed using bcrypt, all production traffic must be encrypted using HTTPS, all user inputs must be validated and sanitized, and API endpoints must have rate limiting to prevent abuse. For reliability, the system must maintain ninety-nine percent uptime during operational hours and have daily automated database backups.

3.4 System Architecture

3.4.1 Architectural Overview

The Bamenda Water & Health Dashboard follows a three-tier client-server architecture consisting of a presentation tier, an application tier, and a data tier. This architectural pattern was selected because it separates concerns, improves maintainability, allows each tier to be scaled independently, and is widely supported by web hosting services. The separation between frontend and backend also enables future mobile application development that could reuse the same API.

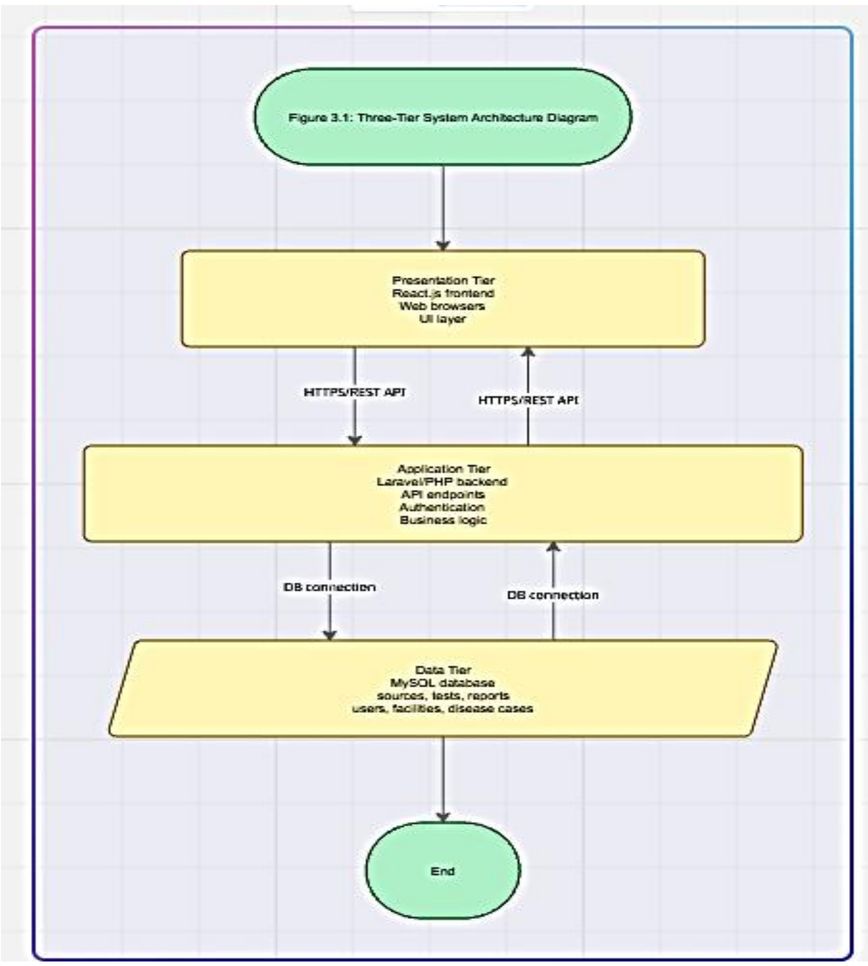
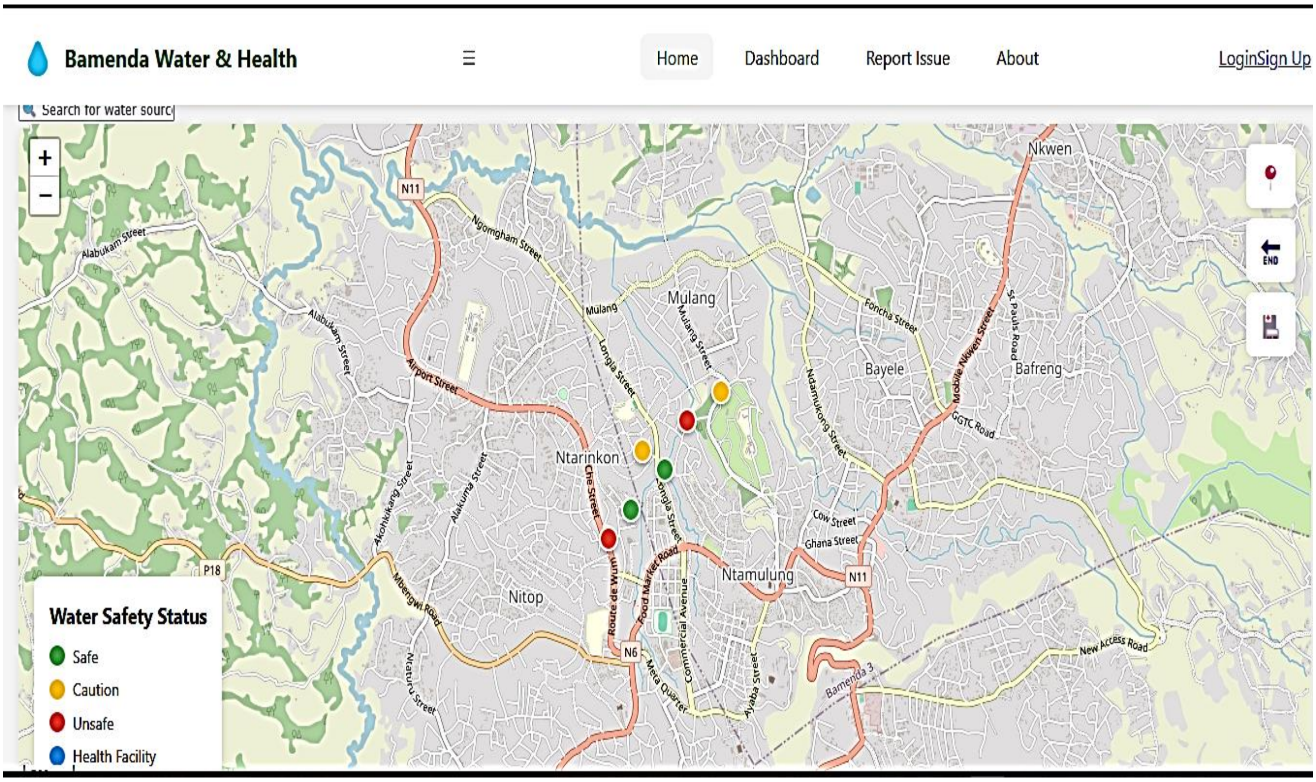


Figure 3.1: Three-Tier System Architecture Diagram showing the Presentation

3.4.2 Presentation Tier (Frontend)

The presentation tier is implemented using standard HTML, CSS, and vanilla JavaScript, chosen for their lightweight nature, compatibility with older devices, and suitability for low-bandwidth connections in Mankon. HTML provides the page structure, CSS handles responsive styling and color-coded status indicators, while JavaScript enables dynamic features such as fetching data from the API, updating the map, and submitting forms without page reloads. Key supporting libraries include Leaflet for interactive mapping, the native Fetch API for HTTP requests, the History API for client-side routing, Chart.js for data visualizations, and Tailwind CSS for accelerated responsive styling. This stack requires no build tools, remains maintainable, and deploys easily on standard web hosting.



3.4.3 Application Tier (Backend)

The application tier handles business logic, authentication, data validation, and API request processing using PHP version 8.1 with the Laravel framework. Laravel was chosen for its robust features including Eloquent ORM for secure database interactions, built-in JWT authentication for session management, validation rules for data integrity, middleware for security filtering, and API resources for consistent JSON responses. The backend exposes a RESTful API organized by resource type. Authentication endpoints manage user registration, login, logout, and password reset. Water sources endpoints support retrieval with filtering and detailed views. Test results endpoints allow adding and retrieving historical test data. Community reports endpoints enable public submission and administrative moderation. Health facilities and disease cases endpoints support health data retrieval and submission, while user management endpoints are restricted to administrators. This RESTful design follows standard HTTP conventions, is stateless for scalability, and is widely documented

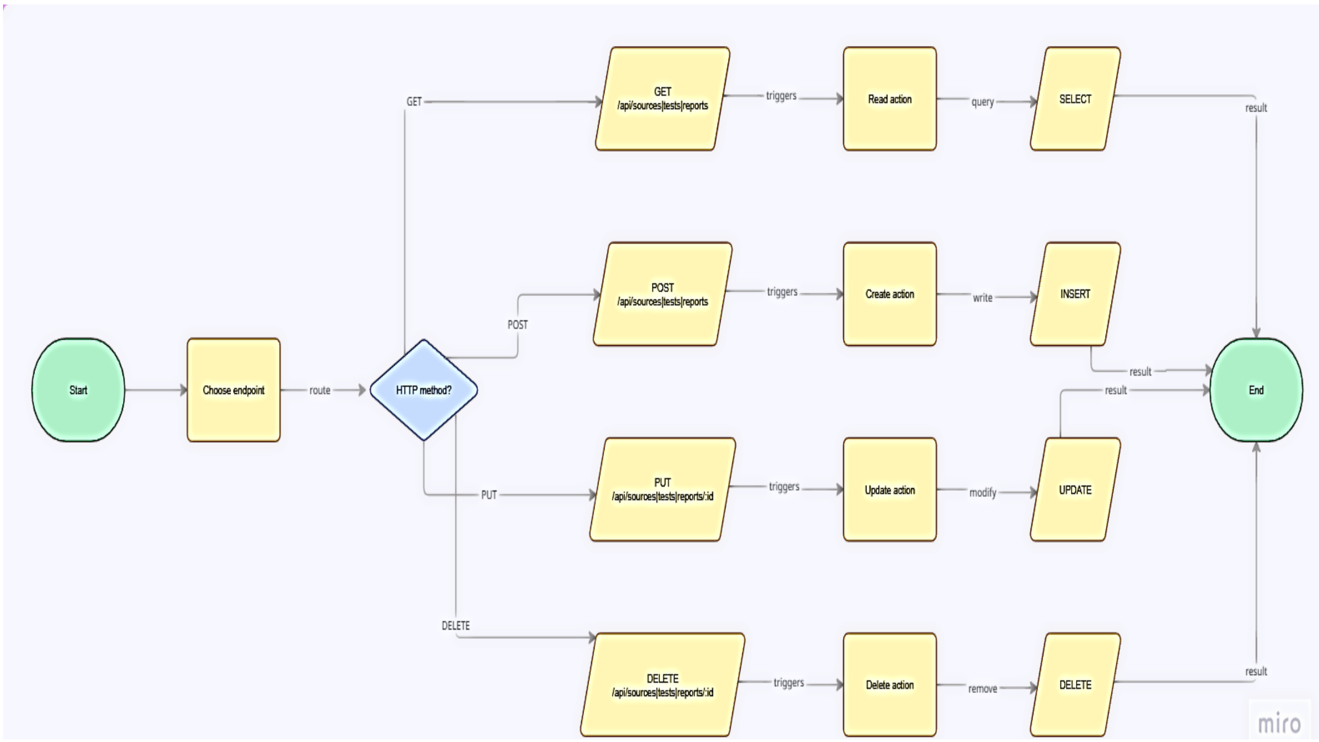


Figure 3.2: API Endpoint Structure Diagram showing the relationship between HTTP methods

3.4.4 Data Tier (Database)

The data tier is implemented using MySQL version 8.0, which runs locally through the XAMPP server environment. XAMPP was chosen as the development server because it provides a complete, ready-to-use web server package that includes Apache, MySQL, PHP, and phpMyAdmin, allowing the database to be easily managed through a graphical interface. MySQL itself is a reliable open-source relational database management system selected for its wide availability on hosting platforms, strong performance for read-heavy workloads typical of web applications, full ACID transaction support for data integrity, and seamless integration with the Laravel framework through Eloquent ORM. The database schema applies key design principles including normalization to third normal form for reduced data redundancy, referential integrity through foreign key constraints, strategic indexing on frequently queried columns for fast response times, and audit fields such as created_at and updated_at timestamps on all tables for tracking data changes.

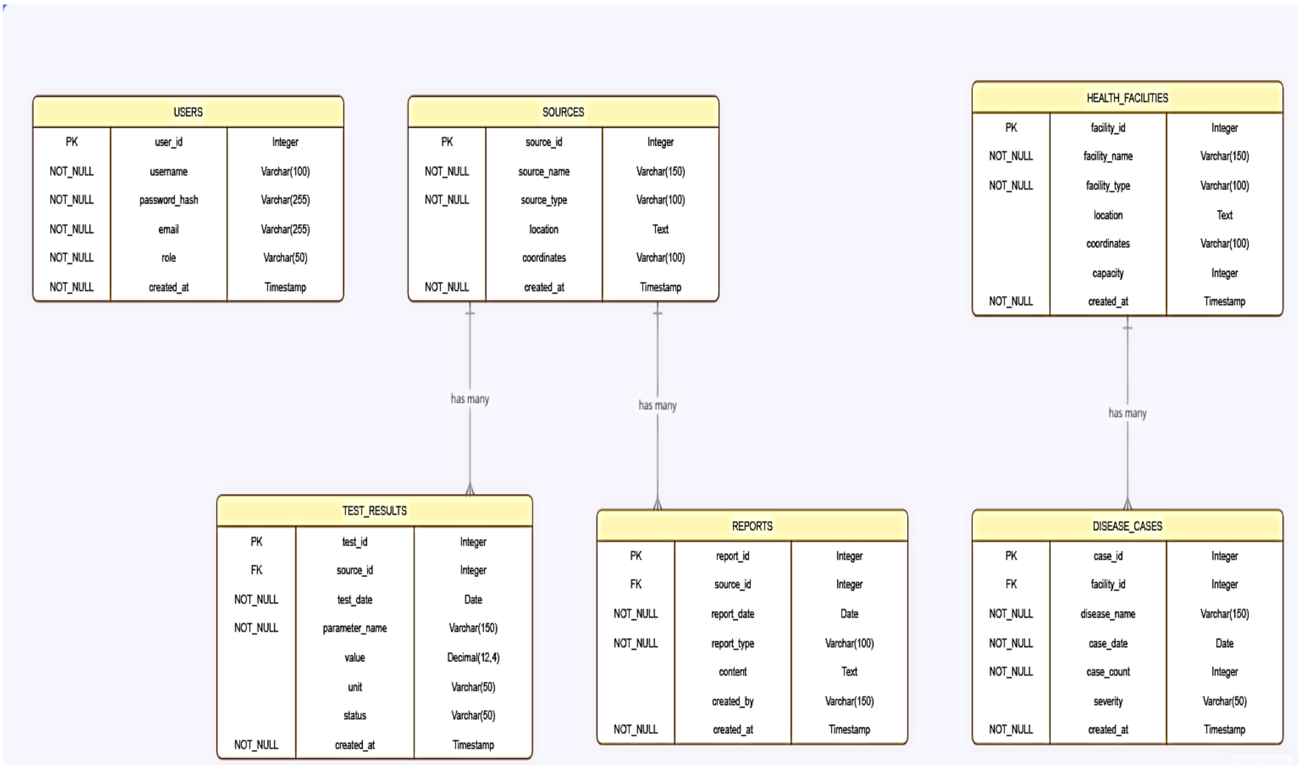


Figure 3.3: Entity-Relationship Diagram showing the six main tables

3.5 Database Design

3.5.1 Table Structures

The database comprises six core tables that support all system functionalities. The users table stores authentication credentials and role assignments including admin, data_entry, health_worker, registered, and anonymous. The sources table captures water source information such as name, GPS coordinates, source type, and current safety status indicated by green, yellow, red, or gray, along with a foreign key linking to the user who created the record. Two tables handle data collection: the test_results table stores official water quality tests with fields for test date, pH level, turbidity, coliform presence, and testing organization, while the reports table captures community observations with a moderation status field tracking pending, approved, or rejected reports.

The remaining two tables support public health integration. The health_facilities table stores locations of clinics and hospitals in Mankon with names, GPS coordinates, and facility types. Linked to this is the disease_cases table, which records weekly statistics for typhoid, diarrhea, and cholera cases per facility. All tables include primary keys, foreign key constraints for referential integrity, and timestamp fields for audit tracking. This relational structure enables the system to connect water quality data with public health outcomes, allowing users to visualize correlations between contaminated water sources and disease incidence in surrounding communities.

3.6 User Interface Design

3.6.1 Design Principles

The user interface design follows five guiding principles. The mobile-first principle prioritizes small screens starting at 320 pixels, then enhances for larger displays using responsive breakpoints, recognizing that smartphones are the primary internet access device in Mankon. The simplicity principle ensures each screen shows only information relevant to the user's current task, with complex features hidden behind progressive disclosure based on role permissions to prevent information overload. The consistency principle applies uniform patterns across the application, including standardized color coding where green indicates safe, yellow indicates caution, red indicates unsafe, gray indicates no data, and blue indicates health-related information, alongside consistent button styles, form layouts, and navigation structures. The feedback principle requires immediate visual confirmation for user actions, such as loading spinners during form submission followed by success or error messages, as well as hover effects and popup animations for map interactions. The accessibility principle ensures color is never the sole means of conveying information, with text labels accompanying colored pins. Touch targets measure at least 44 by 44 pixels, exceeding mobile accessibility recommendations, while the application incorporates ARIA labels and semantic HTML to support screen reader users.

3.6.2 Color Scheme

The color scheme is carefully chosen to communicate meaning intuitively across the application. Green, with hex code 10B981, represents safe water and success messages. Yellow, hex code F59E0B, indicates caution and warnings. Red, hex code EF4444, signifies unsafe conditions, errors, and alerts. Gray, hex code 6B7280, denotes no data or disabled states. Blue, hex code 3B82F6, identifies health facilities and informational content. The primary brand color is a deep green, hex code 059669, used for buttons, links, and active navigation elements. Background color is a light gray, hex code F9FAFB, providing contrast without being harsh on the eyes. Primary text is dark gray, hex code 111827, while secondary text is lighter gray, hex code 6B7280.

	Color Name	Hex Code	Usage Description	Example Application
1	Primary Blue	#0066CC	Main brand color for primary...	Primary buttons, active...
2	Primary Dark Blue	#004C99	Darker variant for hover state...	Button hover states, selected...
3	Secondary Teal	#00A3A3	Secondary brand color for...	Secondary buttons, badges,...
4	Secondary Light Teal	#33BBBB	Lighter variant for backgroun...	Card backgrounds, hover...
5	Success Green	#00CC66	Indicates successful operatio...	Success messages, positive...
6	Success Dark Green	#009950	Darker success variant for...	Success button hover, critical...
7	Warning Orange	#FF9933	Alerts for caution and...	Warning messages, moderate...
8	Warning Amber	#FFAA00	High-priority warnings...	Critical warnings, threshold...
9	Error Red	#E63946	Indicates errors, failures, and...	Error messages, failed...
10	Error Dark Red	#CC2936	Darker error variant for sever...	Critical error states, system...
11	Neutral Gray 100	#F8F9FA	Lightest neutral for...	Page backgrounds, card...
12	Neutral Gray 200	#E9ECEF	Light neutral for borders and...	Borders, dividers, disabled...
13	Neutral Gray 300	#DEE2E6	Medium-light neutral for subt...	Input borders, inactive states,...
14	Neutral Gray 500	#6C757D	Medium neutral for secondar...	Secondary text, captions,...
15	Neutral Gray 700	#495057	Dark neutral for body text	Body text, form labels,...
16	Neutral Gray 900	#212529	Darkest neutral for headings...	Headings, emphasized text,...
17	Accent Purple	#8B5CF6	Accent color for special...	Premium features, special...
18	Accent Cyan	#06B6D4	Accent for data visualization...	Chart elements, data points,...
19	Accent Pink	#EC4899	Accent for notifications and...	Notification badges, alert...
20	White	#FFFFFF	Pure white for maximum...	Text on darkbackgrounds, car...
21	Black	#000000	Pure black for maximum...	Text on light backgrounds,...

Table 3.1: Color Scheme Reference showing each color name, hex code

3.6.3 Screen Designs and Wireframes

The home page serves as the primary interface, featuring an interactive map with color-coded pins for each water source in Mankon, accompanied by a legend, filter controls for safety status and source type, a search box, and a statistics bar showing summary counts. When a user clicks on a pin, the water source detail page appears displaying comprehensive information including a prominent safety status badge, a photo gallery, a test results table showing historical official data in reverse chronological order, a community reports feed of approved observations, and action buttons for submitting new reports or adding official test results for authorized users.

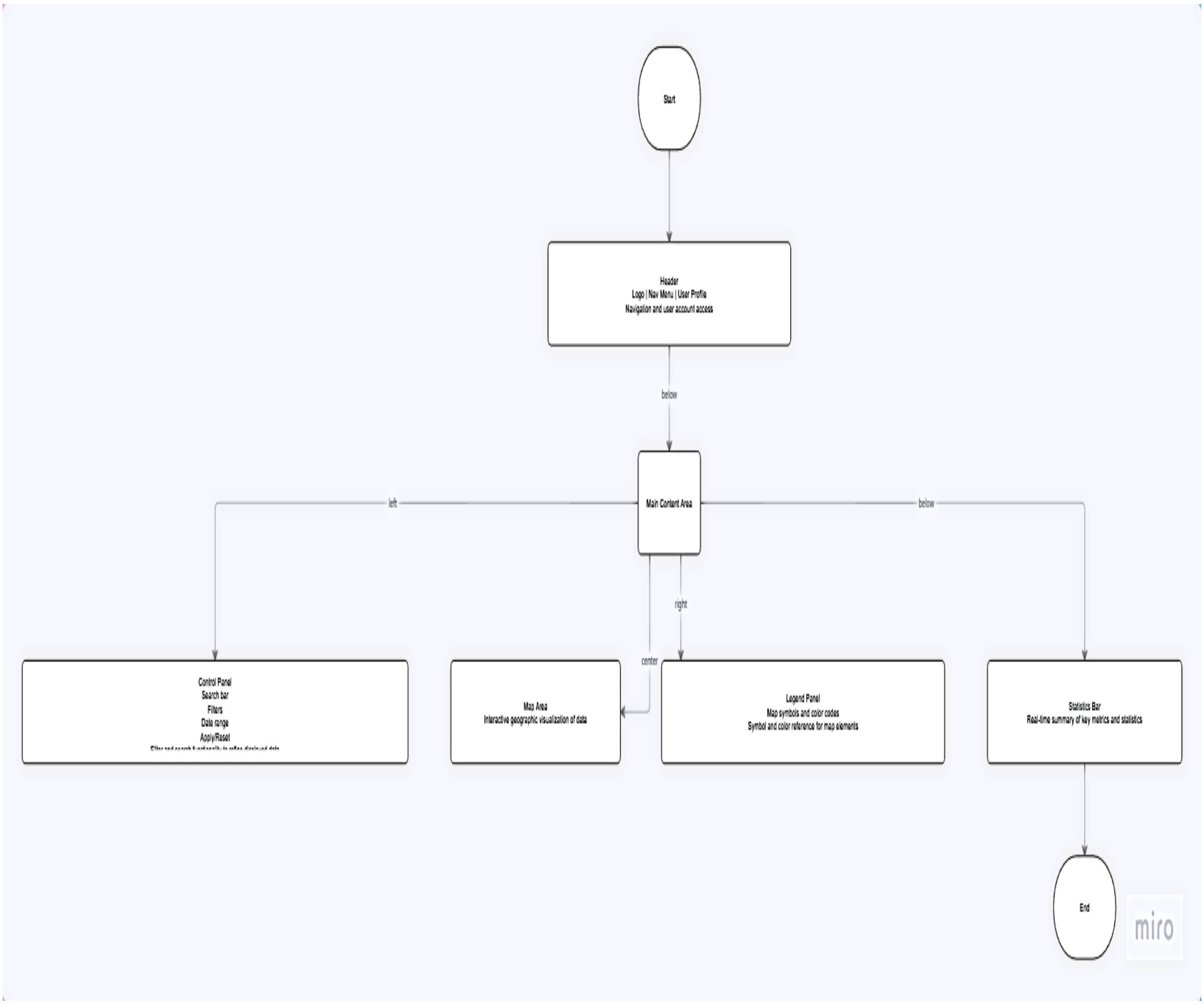


Figure 3.6: Home Page Wireframe showing the layout with header, map area, legend, control panel with filters and search, and statistics bar.

The community report form is a modal overlay with a dropdown for water source selection, icon-based radio buttons for observation type, and optional fields for details, name, and photo upload limited to five megabytes, with only source and observation type required to minimize participation barriers. The admin dashboard uses a tabbed interface organizing management functions into overview, sources, tests, reports, and users tabs, featuring statistics cards, activity logs, source management, test result entry, report moderation with approve and reject buttons, and user account management. Responsive breakpoints ensure adaptation across devices: on mobile below 640 pixels, a single-column layout with the map taking sixty percent of screen height, collapsed accordion legend, card-style tables, and hamburger navigation; on tablets between 640 and 1024 pixels, a two-column layout with compact visible legend; and on desktop above 1024 pixels, a multi-column layout with fully expanded legend and fixed control panel for constant access.

3.7 Processing Logic and Algorithms

3.7.1 Safety Status Calculation Algorithm

The safety status algorithm processes three data sources: the most recent official test result, the age of that test, and recent verified community reports. If no test result exists within the last six months, the status is set to gray indicating no data. If coliform is detected and the test is within three months, the status is set to red indicating unsafe. If coliform is detected but the test is older than three months, the algorithm proceeds to check community reports before a final determination. If no coliform is detected and the test is within three months, the status is provisionally set to green, but if three or more approved community reports from the last thirty days indicate concerning conditions such as cloudy water, bad smell, bad taste, or waste dumping, the status is downgraded from green to yellow. If the test is between three and six months old with no coliform detected, the status is set to yellow indicating caution due to stale data, and if the test is older than six months, the status is set to gray regardless of the result.

Inputs and Assumptions

- Inputs:
  - readings: map of parameterName -> numeric value (e.g., pH, turbidity, lead, arsenic)
  - thresholds: map of parameterName -> {safeMin, safeMax, cautionMin, cautionMax}
- Rules:
  - Safe: value within [safeMin, safeMax]
  - Caution: value within [cautionMin, cautionMax] but outside Safe
  - Unsafe: value outside Caution range or missing thresholds
  - Overall status: Unsafe if any parameter Unsafe; else Caution if any Caution; else Safe

Pseudocode

```
FUNCTION CalculateSafetyStatus(readings, thresholds)
  // Validate inputs
  IF readings IS NULL OR readings IS EMPTY THEN
    RETURN "Unsafe" // No data implies unknown risk; default to Unsafe
  END IF
  overallStatus <- "Safe" // Start optimistic; degrade as needed    FOR EACH param IN
readings.KEYS()       value <- readings[param]       // Skip non-numeric or missing values as Unsafe
for safety     IF value IS NULL OR NOT IS_NUMBER(value) THEN         overallStatus <- "Unsafe"
   CONTINUE       END IF       // Fetch thresholds for this parameter       IF NOT
thresholds.CONTAINS_KEY(param) THEN          // Unknown parameter thresholds -> Unsafe
         overallStatus <- "Unsafe"          CONTINUE       END IF       t <- thresholds[param]       //
Expected structure: t.safeMin, t.safeMax, t.cautionMin, t.cautionMax       IF ANY OF (t.safeMin,
t.safeMax, t.cautionMin, t.cautionMax) IS NULL THEN          overallStatus <- "Unsafe"
CONTINUE       END IF       // Determine per-parameter status       paramStatus <- "Unsafe"       //
Safe range check       IF value >= t.safeMin AND value <= t.safeMax THEN          paramStatus <-
"Safe"       ELSE          // Caution range check (broader than safe)          IF value >= t.cautionMin
AND value <= t.cautionMax THEN             paramStatus <- "Caution"          ELSE
paramStatus <- "Unsafe"          END IF       END IF       // Aggregate overall status using worst-case
logic       IF paramStatus = "Unsafe" THEN          overallStatus <- "Unsafe"          // Early exit:
cannot get worse than Unsafe          BREAK       ELSE IF paramStatus = "Caution" AND overallStatus
= "Safe" THEN          overallStatus <- "Caution"       END IF    END FOR    RETURN overallStatusEND
FUNCTION// Example parameter handling notes:// - pH: safeMin=6.5, safeMax=8.5;
cautionMin=6.0, cautionMax=9.0// - turbidity (NTU): safeMax=1 (safeMin=0), cautionMax=5// -
contaminants (e.g., lead µg/L): safeMax per guideline, cautionMax slightly above; values >
cautionMax -> Unsafe
```

Figure 3.9: Pseudocode for Safety Status Calculation Algorithm showing the step-by-step logic in programming-like syntax.

3.8 Development Methodology

3.8.1 Selection of Agile Methodology

The project follows an agile development methodology with iterative cycles, specifically adopting a simplified version of Scrum suitable for a solo developer working within a bachelor's project timeframe. Agile was selected over traditional waterfall because requirements were expected to evolve as the design progressed, stakeholder feedback was valuable throughout development, and regular working software demonstrations provide better progress visibility than extensive documentation. The short iteration cycles allow for course correction if initial design decisions prove problematic.

3.8.2 Iteration Structure

Each iteration, or sprint, lasts one week and follows a consistent structure. Sprint planning at the beginning of the week selects features from the prioritized backlog based on dependencies and available time. Development occurs throughout the week with daily progress tracking against the plan. Testing is integrated into development, with each feature tested immediately after implementation rather than at the end of the project. A sprint review at the end of the week demonstrates working features, serving as a checkpoint for the project supervisor. A brief retrospective identifies what went well and what could be improved in the next sprint.

3.9 Development Tools and Environment

3.9.1 Hardware Environment

Development was conducted on a laptop computer with an Intel Core i5 processor, sixteen gigabytes of RAM, and a five hundred twelve gigabyte solid-state drive. This configuration provided sufficient performance for running the local development server, database, and frontend build tools simultaneously. Testing was performed on physical mobile devices including an Android smartphone with a six point one inch screen and an iPhone with a five point eight inch screen to verify responsive behavior and touch interactions.

3.9.2 Software Development Tools

The development environment was configured to match production using Ubuntu 22.04 LTS as the operating system, Apache 2.4 as the web server, PHP 8.1 with extensions for PDO database connectivity, OpenSSL encryption, and GD image processing, MySQL 8.0 as the database, and Node.js 18 for frontend build processes. Version control was managed through Git with a private GitHub repository, organized with a main branch for stable code and feature branches merged only after testing completion. Visual Studio Code served as the integrated development environment with extensions for PHP debugging, ESLint for JavaScript quality, Prettier for code formatting, and Tailwind CSS IntelliSense, while TablePlus provided graphical database management for viewing data, running queries, and inspecting table structures.

3.9.3 Technology Stack Summary

Layer	Technology	Version	Justification for Selection
Frontend	HTML5	-	Provides semantic structure for all pages, supports accessibility features (ARIA labels), and ensures cross-browser compatibility
Frontend	CSS3	-	Enables responsive design through media queries, supports modern layouts with Flexbox and Grid, and allows custom styling for color-coded status indicators
Frontend	Vanilla JavaScript	ES6+	Requires no build tools or compilation, works reliably on older devices, reduces bandwidth usage, and is universally supported across all modern browsers
Frontend (Charts)	Chart.js	4.4.2	Simple to implement with vanilla JavaScript, lightweight compared to alternatives, and provides responsive, accessible charts for dashboard visualizations
Frontend (Styling)	Tailwind CSS	3.4.3	Utility-first approach accelerates development, enables responsive design without custom CSS, and purges unused styles for small production bundles
Backend	PHP	8.1	Widely supported on low-cost hosting platforms, has excellent documentation, and integrates seamlessly with MySQL through PDO
Database	MySQL	8.0	Reliable open-source RDBMS with ACID compliance, supports foreign key constraints, has strong performance for read-heavy workloads, and integrates seamlessly with Laravel Eloquent
Development Server	XAMPP	8.2.x	Provides complete local web server environment (Apache, MySQL, PHP, phpMyAdmin) in a single package, simplifying development setup

Version Control	Git	2.40+	Enables change tracking, allows branching for feature development, provides backup through remote repositories, and supports collaboration
IDE	Visual Studio Code	1.85+	Lightweight yet powerful editor with extensive extension ecosystem, built-in terminal, Git integration, and excellent PHP/JavaScript support

Table 3.3: Complete Technology Stack showing each layer (Frontend, Backend, Database, Tools), the specific technology chosen, version number, and justification for selection.

3.10 Ethical Considerations

3.10.1 Data Privacy and Protection

Several ethical considerations guide the design and operation of the system to protect user privacy and ensure responsible data handling. Community report submissions allow anonymity because the name field is optional, encouraging participation from residents who may be concerned about identification or reprisal for reporting problems with community water sources. User accounts store minimal personal information including only name and email address, reducing the potential harm of any data breach. No sensitive health data is collected at the individual level because disease data is aggregated by facility rather than by patient, making re-identification impossible. Users can request deletion of their account and associated data at any time through the account settings page, and such requests are fulfilled within seven days.

3.10.2 Informed Consent

Informed consent is obtained through a consent statement presented during account registration. This statement explains in clear, non-technical language what data is collected, how the data will be used including display on the public map and dashboard, who can access the data including anonymous public users for approved reports, and users' rights regarding their data including access, correction, and deletion. By completing registration, users indicate their understanding and agreement to these terms. For anonymous report submissions that do not require registration, a shorter consent statement is presented above the submit button, explaining that submitted data will be publicly visible after moderation and that no identifying information will be collected unless the user chooses to provide it.

3.10.3 Data Accuracy and Transparency

Data accuracy is promoted through clear labeling of data sources and verification processes. Community reports are clearly labeled as unverified until moderated, with a "pending verification" badge visible to all users. Official test results are clearly distinguished from community reports through separate sections on the water source detail page and through visual styling including a "verified" badge. Simulated demo data used for demonstration purposes is clearly labeled as simulated with a prominent notice on the dashboard, preventing confusion with real data. These transparency measures help users make informed judgments about the reliability of information they see in the system and build trust in the platform.

3.11.1 Use Case Diagram

The use case diagram identifies five external actors and their interactions with the system. Local residents can view the map, view source details, and submit reports. Registered users gain additional dashboard access and report history. Data entry clerks can add test results and manage water sources. Health workers can submit weekly disease case data. Administrators have full system access including user management and report moderation. The diagram shows that viewing and reporting functions are widely available while data management functions are restricted by role.

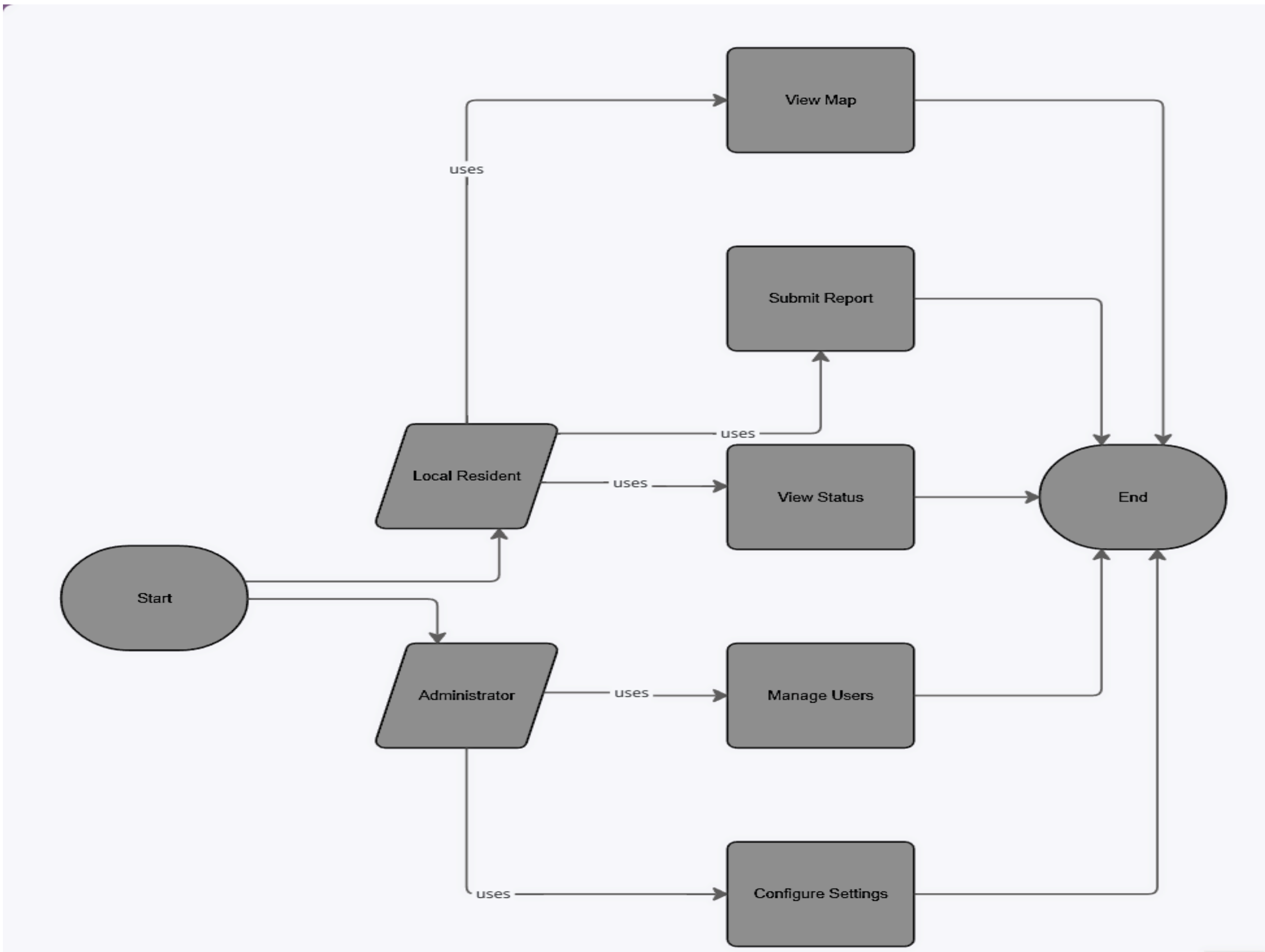


Figure 3.11: Use Case Diagram showing all five actors (Local Resident, Registered User, Data Entry Clerk, Health Worker, Administrator)

3.11.2 Class Diagram

The class diagram represents the static structure of the system with six main classes corresponding to the database tables: User, WaterSource, TestResult, Report, HealthFacility, and DiseaseCase. The diagram shows attributes and methods for each class, along with relationships indicating that one water source can have many test results and many reports, while one health facility can have many disease case records.

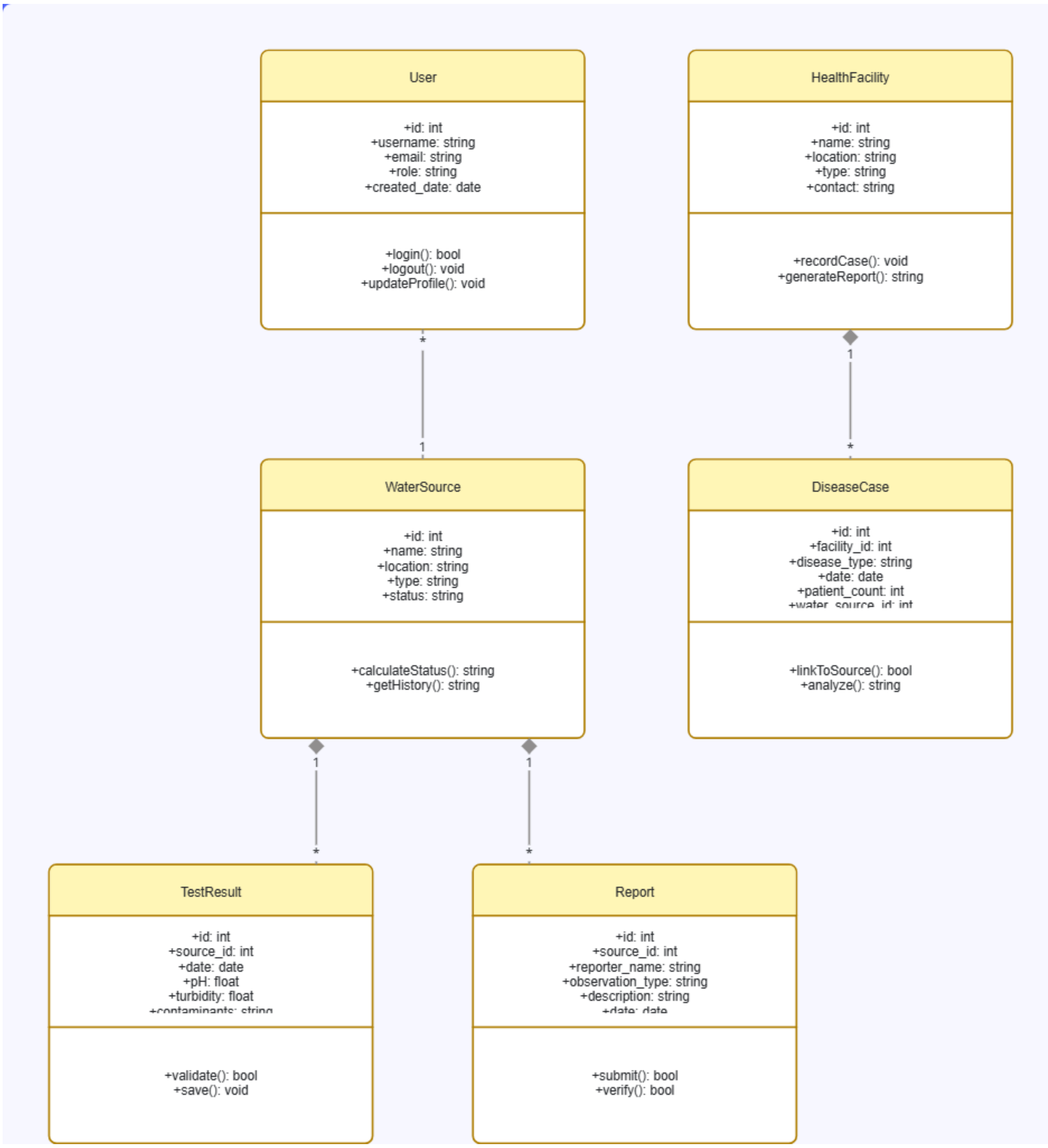


Figure 3.12: Class Diagram showing the six main classes with their attributes and methods, along with association lines indicating relationships and multiplicities

3.11.3 Entity Relationship Diagram

The entity relationship diagram provides a database-level view showing tables as entities, columns as attributes, and foreign key relationships as connections using crow's foot notation. The diagram illustrates that test_results and reports both reference sources, reports optionally references users, disease_cases references health_facilities, and sources references users through the created_by field.

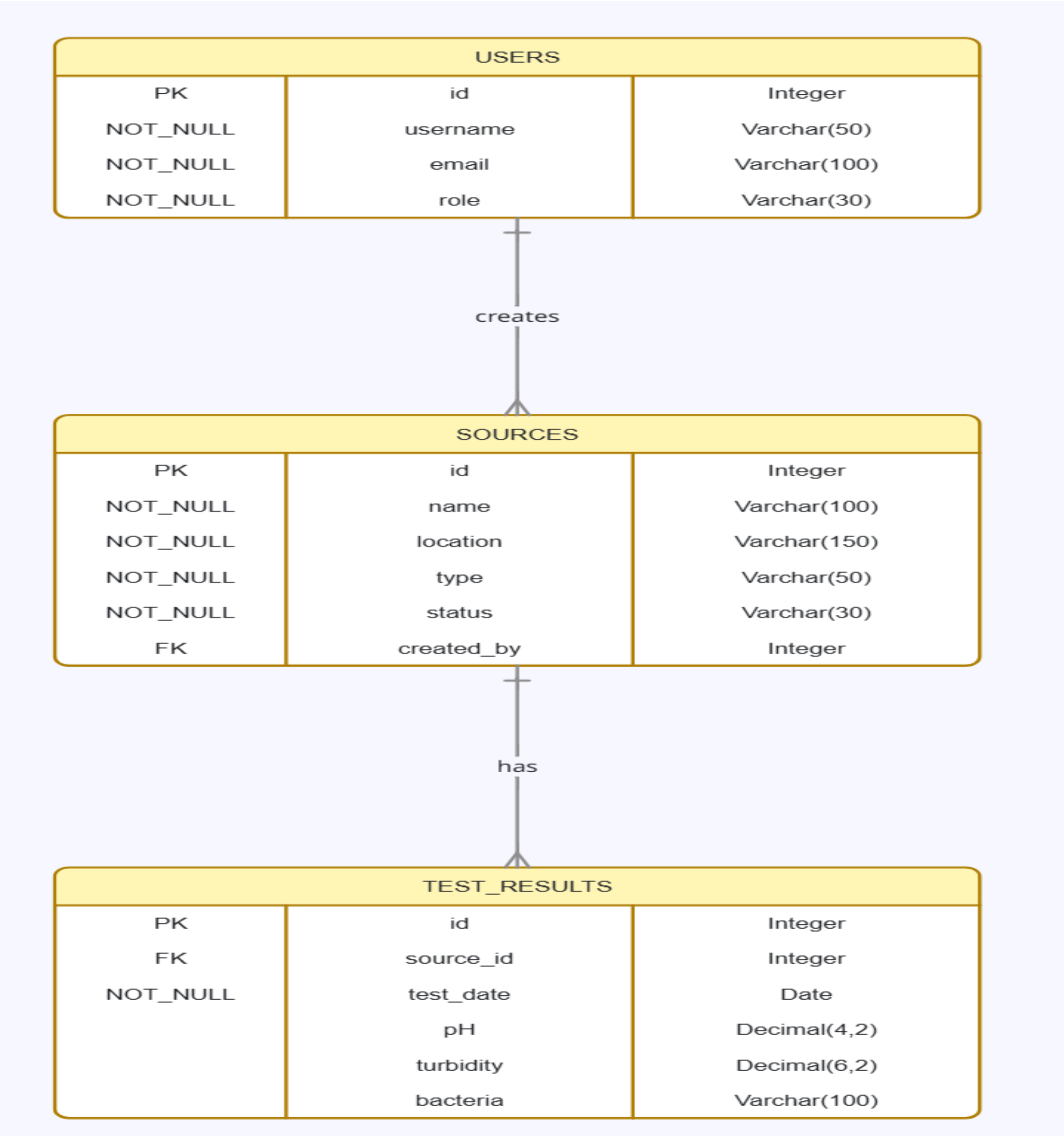


Figure 3.13: Entity Relationship Diagram showing the six tables

3.12 Summary

This chapter has presented the complete research methodology and system design for the Bamenda Water & Health Dashboard. The study adopts a design science research methodology, which is appropriate for creating and evaluating a technological artifact to solve a real-world problem. Requirements were gathered through document analysis, stakeholder analysis, and contextual inquiry, resulting in a comprehensive set of functional and non-functional requirements.

The system follows a three-tier architecture with a React.js frontend for the presentation tier, a Laravel PHP backend for the application tier, and a MySQL database for the data tier. The database design includes six main tables supporting water sources, test results, community reports, users, health facilities, and disease cases, with careful attention to relationships, indexing, and status calculation logic.

The user interface design follows mobile-first principles with color-coded map pins for intuitive communication of water safety status. Wireframes for the home page, detail page, report form, and admin dashboard provide a visual blueprint for implementation. The development methodology is agile with four phases spanning eight weeks. The technology stack is clearly defined with specific versions and justifications. The testing strategy covers unit, integration, system, and user acceptance levels with defined test cases. The deployment plan specifies hosting requirements and step-by-step deployment procedures. Ethical considerations address data privacy, informed consent, and transparency.

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